Homework 8

May 1, 2026

ORF307 Homework 8

Due: Friday, May 1, 2026 9:00 pm ET

- Please export your code with output as pdf.
- If there is any additional answers, please combine them as **ONE** pdf file before submitting to the Gradescope.

Q1 A small integer programming problem

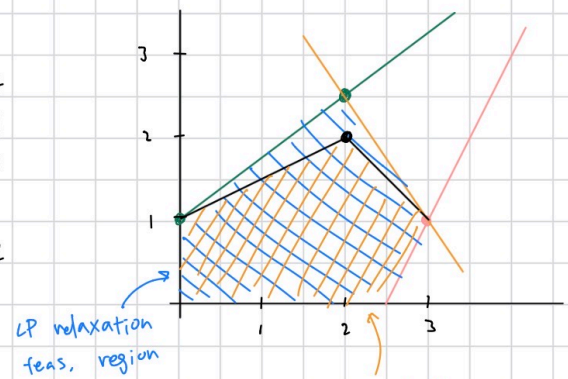
[1]:
```
from IPython.display import Image, display
display(Image("1ab.jpg", width=600))
display(Image("1c.jpg", width=600))
```

Homework 8

① minimize $-x_1 - 2x_2$

subj. to

$$
\begin{aligned}
-3x_1 + 4x_2 &\le 4 \\
3x_1 + 2x_2 &\le 11 \\
2x_1 - x_2 &\le 5 \\
x_1, x_2 &\ge 0 \\
x_1, x_2 &\in \mathbb{Z}
\end{aligned}
$$



1.

Region has vertices:

$(0,0)$, $(0,1)$, $(2, 2.5)$, $(3,1)$, $(2.5, 0)$

↳ minimized $-x_1 - 2x_2 = -2 - 2(2.5) = -7$

so LP optimum is -7 @ $(x_1, x_2) = (2, 2.5)$

Int. Region has feas. points:

$(0,0)$, $(0,1)$, $(1,0)$, $(1,1)$, $(2,0)$, $(2,1)$, $(2,2)$, $(3,1)$

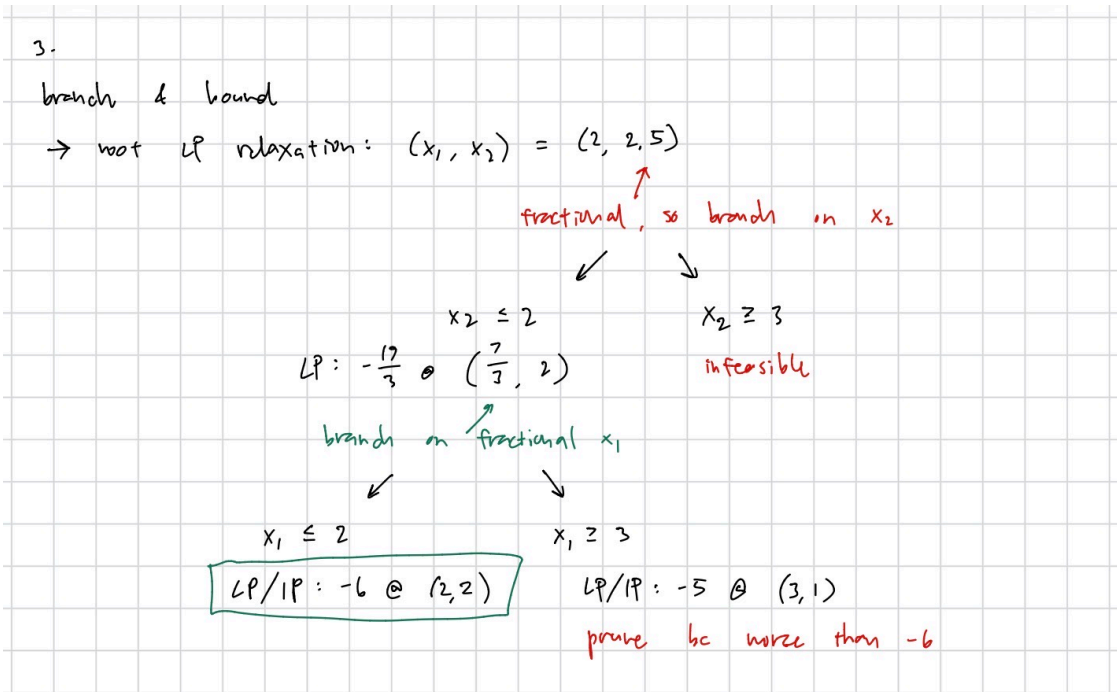
↳ minimized $-x_1 - 2x_2 = -2 - 2(2) = -6$

so integer programming opt. is -6 @ $(x_1, x_2) = (2,2)$

2.

The convex hull of the set of all sols. to the int. programming problem is shaded in orange

It is the region bounded by vertices: $\{(0,0), (0,1), (2,2), (3,1), (2,0)\}$



Q2 Planning to move

It is the end of the semester and you might be planning to change accommodation. Moving is complicated and you want to use some integer programming tricks to make your life easier.

You rented a truck with capacity Q and you bought m boxes. Each box i has size b_i for $i = 1, \dots, m$. You have n items to move of size a_j for $j = 1, \dots, n$.

1. Formulate your packing problem as an integer programming problem to determine if your move is doable with the truck and the boxes you have.
2. Given the data below, is your move feasible? If so, what is the minimum number of boxes you need to use? Answer using the solver `cp.SCIPY`.

[14]:
```
display(Image("2a.png", width=600))
```

(2)

capacity Q : m boxes, n items

↳ sized b_i ↳ sized a_j

Let $x_{ij} = \begin{cases} 1 \; ; & \text{item } j \text{ is in box } i \\ 0 \; ; & \text{o/w} \end{cases}$

$y_i = \begin{cases} 1 \; ; & \text{box } i \text{ is used} \\ 0 \; ; & \text{o/w} \end{cases}$

→ the min-box packing model is

$$\text{minimize} \quad \sum_{i=1}^{m} y_i$$

every item must be assigned to exactly 1 box

$$\text{subj. to} \quad \sum_{i=1}^{m} x_{ij} = 1 \qquad \forall j = 1, \dots, n$$

total size of items in box can't exceed box size

$$\sum_{j=1}^{n} a_j x_{ij} \le b_i y_i \qquad \forall i = 1, \dots, m$$

$$\sum_{i=1}^{m} b_i y_i \le Q$$

total size of boxes can't exceed truck capacity

$$x_{ij}, \; y_i \in \{0, 1\}$$

[7]:
```
import cvxpy as cp
import numpy as np

n = 30
m = 10
Q = 100

a = np.array([4, 3, 1, 2, 2,
              5, 4, 2, 4, 2,
              4, 1, 5, 2, 1,
              3, 5, 3, 3, 3,
              4, 5, 5, 2, 3,
```

```
              3, 3, 2, 2, 2])

b = np.array([7, 19, 10, 14, 10, 12, 13, 10, 15, 14])

# x[i,j] = 1 if item j goes in box i
x = cp.Variable((m, n), boolean=True)

# y[i] = 1 if box i is used
y = cp.Variable(m, boolean=True)

constraints = []

# every item must be assigned to exactly 1 box
constraints.append(cp.sum(x, axis=0) == 1)

# total size of items in box can't exceed box size
constraints.append(x @ a <= cp.multiply(b, y))

# total size of boxes can't exceed truck capacity
constraints.append(b @ y <= Q)



objective = cp.Minimize(cp.sum(y))

prob = cp.Problem(objective, constraints)
prob.solve(solver=cp.SCIPY)

print("Status:", prob.status)
print("Minimum number of boxes:", prob.value)
print("Used boxes:", np.where(y.value > 0.5)[0] + 1)
print("Total truck capacity used:", b @ np.round(y.value))
```

```
Status: optimal
Minimum number of boxes: 7.0
Used boxes: [ 1  2  4  6  7  9 10]
Total truck capacity used: 94.0
```

Q3 Robust production planning

You manage a firm selling two products, with integer production quantities x_1, x_2, subject to the following linear production constraints,

$$\begin{array}{ll} \text{maximize} & c_1 x_1 + c_2 x_2 \\ \text{subject to} & x_1 + x_2 \leq 15 \\ & 2x_1 + x_2 \leq 22 \\ & x_1, x_2 \geq 0 \\ & x_1, x_2 \in \mathbb{Z}. \end{array} \tag{1}$$

You wish to maximize your profit, $c_1x_1 + c_2x_2$, but the per-unit profits c_1, c_2 are uncertain.

The true profit vector c is only known to lie in the box uncertainty set,

$$\mathcal{U} = \{\, c \in \mathbf{R}^2 : |c_i - \bar{c}_i| \le \hat{c}_i,\ i = 1, 2\}, \tag{2}$$

where $\hat{c}$ specifies the half-width of the box in each coordinate, and $\bar{c}$ the nominal/mean profits. This uncertainty set is equivalent to the infinity-norm uncertainty set,

$$\mathcal{U} = \{\, c = Mu + a : \|u\|_\infty \le 1\}, \tag{3}$$

where $M \in \mathbf{R}^{2\times 2}$, and $a \in \mathbf{R}^2$.

[17]:
```
display(Image("3a.png", width=600))
display(Image("3b.png", width=600))
display(Image("3c.png", width=600))
```

(3)

maximize $c_1x_1 + c_2x_2$ subj to

$$x_1 + x_2 \le 15$$
$$2x_1 + x_2 \le 22$$
$$x_1, x_2 \ge 0$$
$$x_1, x_2 \in \mathbb{Z}$$

$\square = \{ c \in \mathbb{R}^2 : |c_i - \bar{c}_i| \le \hat{c}_i,\ i = 1, 2 \}$

↳ each profit coefficient $c_i = \bar{c}_i + \hat{c}_i u_i$ where $-1 \le u_i \le 1$

$\rightarrow c = a + Mu$

so $a = \bar{c}$, $M = \begin{pmatrix} \hat{c}_1 & 0 \\ 0 & \hat{c}_2 \end{pmatrix}$

given the values $\bar{c} = (3.8, 4)$ & $\hat{c} = (0.5, 1)$, we have

$a = \begin{pmatrix} 3.8 \\ 4 \end{pmatrix}$, $M = \begin{pmatrix} 0.5 & 0 \\ 0 & 1 \end{pmatrix}$

b)

robust min. problem w/ ∞-norm uncertainty set

$$\square = \{ c = Mu + a : \|u\|_\infty \le 1 \}$$

max worst case profits = minimize this

Let aux. var t represent the worst-case loss

original uncertain profit $= c_1 x_1 + c_2 x_2 = c^T x$

$\Rightarrow$ worst case loss $= -c^T x =: t$

uncertainty constraint:

$$-(Mu + a)^T x \le t \quad \forall u : \|u\|_\infty \le 1$$

$$\Rightarrow \quad -\left(\begin{pmatrix} 0.5 & 0 \\ 0 & 1 \end{pmatrix} u + \begin{pmatrix} 3.8 \\ 4 \end{pmatrix} \right)^T \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \le t$$

$$\Rightarrow \quad -(3.8 + 0.5u_1) x_1 - (4 + u_2) x_2 \le t \qquad \forall \|u_1\| \le 1, \ \|u_2\| \le 1$$

worst case loss $\rightarrow$ smallest profits

$$\Rightarrow \quad c_1 = 3.8 - 0.5 = 3.3, \qquad c_2 = 4 - 1 = 3$$

so the robust min problem becomes:

$$\text{minimize}: \ t := -c^T x, \qquad \text{subj to}: \ \left.\begin{array}{l} x_1 + x_2 \le 15 \\ 2x_1 + x_2 \le 22 \\ x_1, x_2 \ge 0 \\ x_1, x_2 \in \mathbb{Z} \end{array}\right\} \text{original constraints}$$

robust uncertainty constraint $\rightarrow$ $-3.3x_1 - 3x_2 \le t$

(c)

robust opt: minimize $(Mu + a)^T x = a^T x - \|M^T x\|_1$

deterministic rob.: $\underset{(x)}{\text{maximize}}$ $a^T x - \|M^T x\|_1$

subject to
$$\left.\begin{aligned} x_1 + x_2 &\le 15 \\ 2x_1 + x_2 &\le 22 \\ x_1, x_2 &\ge 0 \\ x_1, x_2 &\in \mathbb{Z} \end{aligned}\right\} (*)$$

$\rightarrow$ $a = \begin{pmatrix} 3.8 \\ 4 \end{pmatrix}$, $M = \begin{pmatrix} 0.5 & 0 \\ 0 & 1 \end{pmatrix}$

$\rightarrow$ maximize $(3.8, 4)\, x - \left\| \begin{pmatrix} 0.5 & 0 \\ 0 & 1 \end{pmatrix} x \right\|_1$

$\rightarrow$ since $x_1, x_2 \ge 0$, this becomes: max $3.3x_1 + 3x_2$

subj to $(*)$

[18]:
```python
import cvxpy as cp
import numpy as np

a = np.array([3.8, 4.0])
M = np.array([[0.5, 0.0],
              [0.0, 1.0]])

x = cp.Variable(2, integer=True)

objective = cp.Maximize(a @ x - cp.norm(M.T @ x, 1))

constraints = [
    x[0] + x[1] <= 15,
    2*x[0] + x[1] <= 22,
    x >= 0
]

prob = cp.Problem(objective, constraints)
prob.solve(solver=cp.SCIPY)
```

```
print("Status:", prob.status)
print("Robust optimal worst-case profit:", prob.value)
print("x:", np.round(x.value))
```

```
Status: optimal
Robust optimal worst-case profit: 47.099999999999994
x: [7. 8.]
```

[19]:
```
display(Image("3d.png", width=600))
```

ⓓ This uses $c = \bar{c} = a$, so the problem becomes

maximize $3.8x_1 + 4x_2$

subj. to (*) the same constraints

.....

[20]:
```
x_nom = cp.Variable(2, integer=True)

objective_nom = cp.Maximize(a @ x_nom)

constraints_nom = [
    x_nom[0] + x_nom[1] <= 15,
    2*x_nom[0] + x_nom[1] <= 22,
    x_nom >= 0
]

prob_nom = cp.Problem(objective_nom, constraints_nom)
prob_nom.solve(solver=cp.SCIPY)

print("Status:", prob_nom.status)
print("Nominal optimal profit:", prob_nom.value)
print("x:", np.round(x_nom.value))
```

```
Status: optimal
Nominal optimal profit: 60.0
x: [-0. 15.]
```

so, the robust solution is: x = (7, 8) with worst-case profit 47.1

the nominal solution is: x = (0, 15) with nominal profit 60

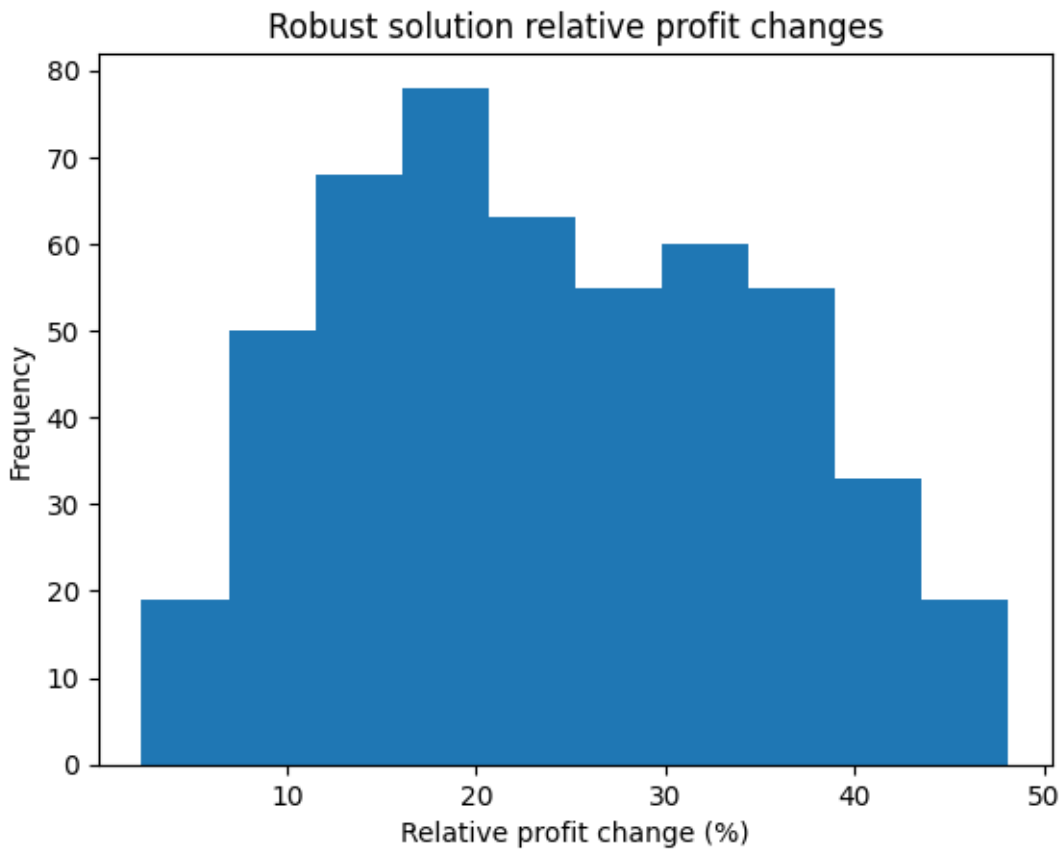
however, under worst-case profits, the nominal solution only earns $3.3(0) + 3(15) = 45$

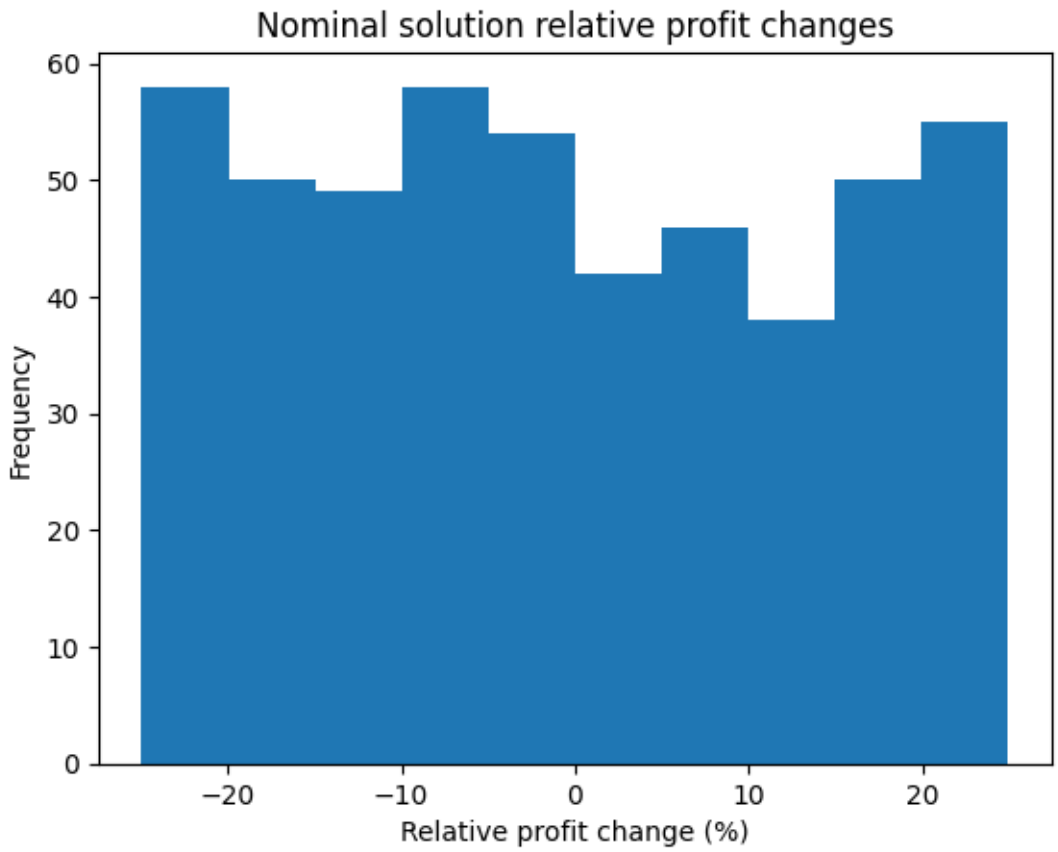
so, the nominal solution has a higher nominal profit, but the robust solution is better under the worst-case.

[21]:
```
import numpy as np
import matplotlib.pyplot as plt

np.random.seed(0)

c_bar = np.array([3.8, 4])
c_hat = np.array([0.5, 1.0])

D = np.random.uniform(c_bar - c_hat, c_bar + c_hat, size=(500, 2))

x_robust = np.array([7, 8])
x_nominal = np.array([0, 15])

pstar_robust = 47.1
pstar_nominal = 60.0

# profits for each sample
profits_robust = D @ x_robust
profits_nominal = D @ x_nominal

print("Robust solution x = (7,8)")
print("Average profit:", np.mean(profits_robust))
print("Standard deviation:", np.std(profits_robust))
print("Worst-case dataset profit:", np.min(profits_robust))

print()

print("Nominal solution x = (0,15)")
print("Average profit:", np.mean(profits_nominal))
print("Standard deviation:", np.std(profits_nominal))
print("Worst-case dataset profit:", np.min(profits_nominal))

# relative profit changes
rel_robust = 100 * (profits_robust - pstar_robust) / pstar_robust
rel_nominal = 100 * (profits_nominal - pstar_nominal) / pstar_nominal

plt.hist(rel_robust)
plt.xlabel("Relative profit change (%)")
plt.ylabel("Frequency")
plt.title("Robust solution relative profit changes")
plt.show()

plt.hist(rel_nominal)
plt.xlabel("Relative profit change (%)")
```

```
plt.ylabel("Frequency")
plt.title("Nominal solution relative profit changes")
plt.show()
```

```
Robust solution x = (7,8)
Average profit: 58.44349297827592
Standard deviation: 5.13992322593276
Worst-case dataset profit: 48.20996618486927

Nominal solution x = (0,15)
Average profit: 59.66863832356986
Standard deviation: 8.816674420336595
Worst-case dataset profit: 45.01637894690987
```





the nominal solution has a slightly higher average profit because it chooses x = (0, 15), which puts everything into product 2 (which has higher nominal profit at c_2 = 4)

however, product 2 has larger uncertainty (1) than product 1 (0.5) so the nominal solution is more risky. this makes its standard deviation much larger and the worst-case profit worse than that of the robust solution

the robust solution @ x = (7, 8) is more steady and balanced. it has less average profit but has lower variability and a better worst-case profit

[22]:
```
display(Image("3f.png", width=600))
```

f)

The LP relaxation removes the int. constraint $(x_1, x_2 \in \mathbb{Z})$

$\rightarrow$ now: $x_1, x_2 \geq 0$ can be continuous vars

$\Rightarrow$ nominal LP relaxation becomes

$$\begin{aligned} \text{maximize:} \quad & 3.8x_1 + 4x_2 \\ \text{subj. to:} \quad & x_1 + x_2 \leq 15 \\ & 2x_1 + x_2 \leq 22 \\ & x_1, x_2 \geq 0 \end{aligned}$$

[23]:
```python
import cvxpy as cp
import numpy as np

x_lp = cp.Variable(2)

objective = cp.Maximize(3.8*x_lp[0] + 4*x_lp[1])

constraints = [
    x_lp[0] + x_lp[1] <= 15,
    2*x_lp[0] + x_lp[1] <= 22,
    x_lp >= 0
]

prob = cp.Problem(objective, constraints)
prob.solve(solver=cp.SCIPY)

print("Status:", prob.status)
print("LP relaxation optimal profit:", prob.value)
print("x:", x_lp.value)
```

```
Status: optimal
LP relaxation optimal profit: 60.0
x: [-0. 15.]
```

this is just the same integer solution as we got in part d. this is because the LP optimum is already at an integer vertex of the feasible region

so, relaxing the integer constraint has no effect and doesn't chance our solution